

A Curriculum-Aware Adaptive Academic Planning System using custom weighted decision logic for intelligent course priority allocation and risk evaluation.

Frontend Engineering

React & TypeScript v19.2 / v5.9

Core web framework enabling structured, type-safe, component-driven user interface development.

Vite v7.3

Next-generation frontend tooling providing ultra-fast Hot Module Replacement and production bundling.

Tailwind CSS & PostCSS v4.2

Utility-first CSS framework with native CSS post-processing for rapid, modern responsive layouts.

React Router Dom v7.13

Client-side declarative routing solution for multi-page application state management.

Recharts v3.7

Composable charting library built on React components for rendering performance & analytics data.

Framer Motion v12.34

Production-ready motion library for fluid UI state transitions and micro-animations.

Backend Architecture

Node.js & Express v5.2 (Express)

Asynchronous, event-driven server runtime hosting the RESTful API endpoints and application logic.

Google Generative AI (Gemini) v0.24

Integrates Gemini model APIs to generate personalized academic insights and explanation metrics.

Prisma ORM v5.22

Type-safe database ORM managing schema definitions, migrations, and clean programmatic queries.

Authentication (JWT / bcryptjs) v9.0 / v3.0

Stateless token-based authentication paired with secure salted password hashing algorithms.

pdf-parse v2.4

Server-side utility for extracting text content from uploaded syllabus and curriculum PDFs.

Database & Core Engines

PostgreSQL Database pg v8.18

Relational database storage containing user profiles, selected curricula, weekly availabilities, study plan sessions, and quizzes.

Academic Priority & Risk Engines Custom Logic

Backend engines resolving PriorityScore and RiskFactor based on credits, course difficulty, exam countdowns, and weekly study hours.

Tooling & Deploy

Vercel Integration vercel.json

Deployment routing and backend serverless function configurations.

Concurrently / Nodemon v8.2 / v3.1

Development script tools ensuring quick hot-reloading for both servers simultaneously.